

8 Statistics

8.1 Populations and Samples

“There are three kinds of lies – lies, damned lies, and statistics.”

(British Prime Minister Benjamin Disraeli)

Definition 8.1.1 The **population** of study is the group the collected data is intended to describe.

It is important to specify the population because our conclusions made from the data could differ based on the population we wish to study.

Sometimes known as the target population, since a poorly designed study could fail to represent the true population.

Example 8.1.2 Suppose we wish to study the average amount of money spent on textbooks for first-year college students. The population we wish to study might be any of the following:

- First year students at the University of Idaho.
- First year students at any university in Idaho.
- First year students at any community college in Idaho.
- First year students at any community college in the USA.
- Etc.

Maybe there is a class at the university of Idaho which requires a particularly expensive textbook, which could skew our conclusion about the population if the target population is first year students at universities in the USA.

Definition 8.1.3 If we are able to gather data on every single member of the population, any value computed using all the data is called a **parameter**.

Definition 8.1.4 A survey of the entire population is called a **census**

You are probably familiar with a few common censuses: the United States Census, and voting.

voting in particular shows the difficulty in surveying the entire population: some may not answer the survey.

Definition 8.1.5 A **sample** is a subset of the population, ideally one that is a good representation of the population.

Example 8.1.6 A researcher wanted to know how citizens of Tacoma felt about a voter initiative. To study this, she goes to the Tacoma Mall and randomly selects 500 shoppers and asks them their opinion. 60% indicate they are supportive of the initiative. What is the sample and population? Is the 60% value a parameter or a statistic?

Note that the effective population was mall shoppers, and there is no reason to believe that this is a representative sample of the target population.

Categorizing Data

Definition 8.1.7 **Categorical (qualitative) data** are pieces of information that allow us to classify the objects under investigation into categories.

Quantitative data are responses that are numerical in nature that allow us to perform meaningful calculations.

Example 8.1.8 Suppose we put out a survey asking how many movies people have watched in the last year. What kind of data is this?

Example 8.1.9 Suppose we ask people what ZIP code they live in. What kind of data is this?

Sampling Methods

Example 8.1.10 Suppose we are hired by a politician to determine how much support they would have if they decided to run for another term? As always, the first step is identifying the population. The next step is to take a sample of the population. How do we do this?

All people in area? All voters? All registered voters? All likely voters? This last one is what is commonly used, but it can still fail.

In 1998, former professional wrestler Jesse Ventura was elected governor of Minnesota. Polls before the election consistently showed that he had little chance of winning. One big factor of why this happened was because polls omitted younger voters, since they were not considered likely voters. Since he had a lot of support in this group, the polls were not accurate.

This kind of situation is known as sampling bias, and it can be difficult to avoid. In this example, the polls omitted a key group from their data, and so the sample was not a good representation of the target population.

Definition 8.1.11 A sampling method is called **biased** if every member of the population doesn't have equal likelihood of being in the sample.